

LECTURES 3 - 4

Theorem 0.1 (*Properties of probability measure*) Let $(\Omega, \mathcal{F}, P)$ a probability space and $A, B, A_1, A_2, \dots$ are in $\mathcal{F}$. Then

(1) $P(A^c) = 1 - P(A)$.

(2) *Monotonicity:* if $A \subseteq B$, then

$$P(A) \leq P(B).$$

(3) *Inclusion - exclusion formula:*

$$\begin{aligned} P\left(\bigcup_{k=1}^n A_k\right) &= \sum_{k=1}^n P(A_k) - \sum_{1 \leq i < j \leq n} P(A_i \cap A_j) \\ &\quad + \sum_{1 \leq i < j < k \leq n} P(A_i \cap A_j \cap A_k) + \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n). \end{aligned}$$

(4) *Finite sub-additivity:*

$$P(A \cup B) \leq P(A) + P(B).$$

(5) *Continuity property:*

(i) For $A_1 \subseteq A_2 \subseteq \dots$

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n).$$

(ii) For $A_1 \supseteq A_2 \supseteq \dots$,

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n).$$

(6) *Boole's inequality (Countable sub-additivity):*

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} P(A_n).$$

Proof. Since $\Omega = A \cup A^c$, using finite additivity, we have

$$1 = P(A) + P(A^c).$$

This implies (1).

Let

$$A \subseteq B, \text{ then } B = A \cup (B \setminus A).$$

Therefore

$$P(B) = P(A) + P(B \setminus A) \Rightarrow P(B) \geq P(A),$$

since $P(B \setminus A) \geq 0$. This proves (2).

We prove (3) by induction. For $n = 2$

$$A_1 \cup A_2 = A_1 \cup (A_2 \setminus A_1).$$

and

$$A_2 = (A_2 \setminus A_1) \cup (A_1 A_2).$$

(Here $A_1 A_2 = A_1 \cap A_2$). Hence we have

$$P(A_1 \cup A_2) = P(A_1) + P(A_2 \setminus A_1)$$

and

$$P(A_2) = P(A_2 \setminus A_1) + P(A_1 A_2).$$

Combining the above, we have

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 A_2).$$

Assume that equality holds for $n \leq m$. Consider

$$\begin{aligned} P(\cup_{k=1}^{m+1} A_k) &= P(\cup_{k=1}^m A_k \cup A_{m+1}) \\ &= P(\cup_{k=1}^m A_k) + P(A_{m+1}) - P((\cup_{k=1}^m A_k) \cap A_{m+1}) \\ &= P(\cup_{k=1}^m A_k) + P(A_{m+1}) - P(\cup_{k=1}^m A_k A_{m+1}) \\ &= \sum_{k=1}^m P(A_k) - \sum_{1 \leq i < j \leq m} P(A_i A_j) \\ &\quad + \cdots + (-1)^{m+1} P(A_1 \cdots A_m) + P(A_{m+1}) \\ &\quad - \left[\sum_{k=1}^m P(A_k A_{m+1}) - \sum_{1 \leq i < j \leq m} P(A_i A_{m+1} A_j A_{m+1}) \right] \\ &\quad + \cdots + (-1)^{m+1} P(A_1 A_{m+1} A_2 A_{m+1} \cdots A_m A_{m+1}) \end{aligned}$$

$$\begin{aligned}
&= \sum_{k=1}^{m+1} P(A_k) - \left[\sum_{1 \leq i < j \leq m} P(A_i A_j) + \sum_{k=1}^m P(A_k A_{m+1}) \right] \\
&\quad + \left[\sum_{1 \leq i < j < k \leq m} P(A_i A_j A_k) + \sum_{1 \leq i < j \leq m} P(A_i A_j A_{m+1}) \right] \\
&\quad + \cdots + (-1)^{m+2} P(A_1 A_2 \cdots A_m A_{m+1}) \\
&= \sum_{k=1}^{m+1} P(A_k) - \sum_{1 \leq i < j \leq m+1} P(A_i A_j) \\
&\quad + \sum_{1 \leq i < j < k \leq m+1} P(A_i A_j A_k) \cdots + (-1)^{m+2} P(A_1 \cdots A_{m+1}).
\end{aligned}$$

Therefore the result true for $n = m + 1$. Hence by induction property (3) follows.

From property (3), we have

$$P(A \cup B) = P(A) + P(B) - P(AB).$$

Hence

$$P(A \cup B) \leq P(A) + P(B).$$

Thus we have (4).

Now we prove (5)(i). Set

$$B_1 = A_1, B_n = A_n \setminus A_{n-1}, n = 2, 3, \dots$$

Then

$$B_n \in \mathcal{F} \quad \forall n = 1, 2, \dots,$$

B'_n s are pairwise disjoint and

$$A_n = \cup_{k=1}^n B_k, n \geq 1. \quad (0.1)$$

Also

$$\cup_{n=1}^{\infty} B_n = \cup_{n=1}^{\infty} A_n. \quad (0.2)$$

To see the above,

$$\begin{aligned}
\omega \in \cup_{n=1}^{\infty} A_n &\Rightarrow \omega \in A_n = \cup_{k=1}^n B_k \text{ for some } n \geq 1 \\
&\Rightarrow \omega \in B_m \text{ for some } m \leq n \\
&\Rightarrow \omega \in \cup_{n=1}^{\infty} B_n.
\end{aligned}$$

From $B_n \subseteq A_n, n \geq 1$, reverse inclusion follows.

Using (0.1), we get

$$P(A_n) = \sum_{k=1}^n P(B_k), n \geq 1.$$

Hence, from the definition of series convergence, we get

$$\lim_{n \rightarrow \infty} P(A_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n P(B_k) = \sum_{n=1}^{\infty} P(B_n). \quad (0.3)$$

Similarly, using (0.2), we get

$$P(\cup_{n=1}^{\infty} A_n) = P(\cup_{n=1}^{\infty} B_n) = \sum_{n=1}^{\infty} P(B_n).$$

Therefore from (0.3), we have

$$P(\cup_{n=1}^{\infty} A_n) = \lim_{n \rightarrow \infty} P(A_n).$$

Proof of (5)(ii) is as follows.

Using

$$A_1^c \subseteq A_2^c \cdots$$

and (5)(i) we get (exercise)

$$\lim_{n \rightarrow \infty} P(A_n) = P(\cap_{n=1}^{\infty} A_n).$$

From property (4) and principle of mathematical induction, it follows that

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) \leq P(A_1) + \cdots + P(A_n) \quad \forall n \geq 1.$$

i.e.,

$$P(\cup_{k=1}^n A_k) \leq \sum_{k=1}^n P(A_k) \quad \forall n \geq 1.$$

Therefore

$$P(\cup_{k=1}^n A_k) \leq \sum_{k=1}^{\infty} P(A_k) \quad \forall n \geq 1. \quad (0.4)$$

Set

$$B_n = \cup_{k=1}^n A_k.$$

Then $B_1 \subseteq B_2 \subseteq \dots$ and are in $\mathcal{F}$. Also

$$\cup_{n=1}^{\infty} A_n = \cup_{n=1}^{\infty} B_n.$$

Hence

$$P(\cup_{n=1}^{\infty} A_n) = P(\cup_{n=1}^{\infty} B_n) = \lim_{n \rightarrow \infty} P(B_n) = \lim_{n \rightarrow \infty} P(\cup_{k=1}^n A_k). \quad (0.5)$$

Here the second equality follows from the continuity property 5(i). Using (0.5), letting $n \rightarrow \infty$ in (0.4), we have

$$P(\cup_{n=1}^{\infty} A_n) \leq \sum_{k=1}^{\infty} P(A_k).$$

Let us end this chapter with the formulation and solution to the points problem which is insprit the solution given by Pascal initiated by Fermat.

(Reading exercise) Note since I need m points and II need l points, in the worst case one need $m+l-1$ to complete the game. Hence consider the random experiment of $m+l-1$ tosses of a fair coin (there is a missing link that none of the tosses influence each other. This will be clarified at a later part of the course) This gives Ω as

$$\Omega = \{(\omega_1, \dots, \omega_{m+l-1}) | \omega_i \in \{H, T\}\}, \mathcal{F} = \mathcal{P}(\Omega), P(A) = \frac{\#A}{\#\Omega}.$$

Note that $\#\Omega = 2^{m+l-1}$.

Let E denote the event that I gets atleast m points. Then E^c is the event that II gets atleast l . Then

$$\begin{aligned} \#E &= \binom{m+l-1}{m} + \binom{m+l-1}{m+1} + \dots + \binom{m+l-1}{m+l-1} \\ \#E^c &= \binom{m+l-1}{0} + \binom{m+l-1}{1} + \dots + \binom{m+l-1}{m-1}. \end{aligned}$$

In the last equality, we use

$$\binom{a}{b} = \binom{a}{a-b}, \quad a, b \in \mathbb{N}, \quad b \leq a.$$

This gives the probabilities p and q for the win of A and B respectively. Hence clearly

$$\frac{p}{q} = \frac{\#E}{\#E^c}.$$

This gives the division of the prize money $p : q$

Now I will do an exercise in detail to illustrate how to solve problems. This is intended as an illustration of 'how to approach a problem'

Problem 1: Let $\mathcal{F}$ be a field with finitely many elements. Show that $\mathcal{F}$ is a σ -field.

Solution-explained: By comparing the definitions of field and σ -field, it is evident that one need to show the following "If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$."

One may immediately notice that "since there are only finitely many elements in $\mathcal{F}$, any countable union is going to be a finite union". Intuition is correct but how do one establish this? One may say, it is 'trivial' because A_n are taken from a finite collection, so A_n 's should repeat after some n onwards. Convincing but some thing is missing in the 'proof', in fact it is too vague.

I will give a proof, using set theory we know. From given, there exists n such that $\#\mathcal{F} = n$. Hence $\mathcal{F} = \{B_1, B_2, \dots, B_n\}$ for some $B_i \in \mathcal{P}(\Omega), i = 1, 2, \dots, n$. Now since $A_n \in \mathcal{F}$ for all n , we have $\{A_1, A_2, \dots\} \subseteq \{B_1, B_2, \dots, B_n\}$ (definition of subset). Hence there exists $\{i_1, i_2, \dots, i_k\} \subseteq \{1, 2, \dots, n\}$ such that $\{A_1, A_2, \dots\} = \{B_{i_1}, B_{i_2}, \dots, B_{i_k}\}$. This is what one meant by A_n 's repeat after some time (no ambiguity here). Hence

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{i=1}^k B_{i_i}.$$

i.e. the proof of countable union is indeed a finite union is complete. Now $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$, since $\mathcal{F}$ is a field.

Problem 2: Let $\{A_1, A_2, \dots, A_n\}$ be a partition of Ω . Show that $\sigma(\{A_1, A_2, \dots, A_n\})$ is the set of all finite union of A_i 's.

Solution-explained Set

$$\mathcal{D} = \text{set of all finite union of } A_i' \text{'s.}$$

Our task to show that $\sigma(\{A_1, A_2, \dots, A_n\}) = \mathcal{D}$.

By looking at the definition of $\sigma(\{A_1, A_2, \dots, A_n\})$, we need to show that

1. $\mathcal{D}$ is a σ -field,
2. $A_1, A_2, \dots, A_n \in \mathcal{D}$,

3. $\mathcal{D}$ is the smallest σ -field satisfying 2. (In other words, if $\mathcal{G}$ is a σ -field and $\{A_1, A_2, \dots, A_n\} \subseteq \mathcal{G}$, then $\mathcal{D} \subseteq \mathcal{G}$.)

Once roughly the tasks are identified, it is about completing these tasks. Note that first two tasks, i.e. 1 and 2 are just verifications.

For checking 1., we need to verify the definition of σ -field for $\mathcal{D}$. Let us do this.

Note $\Omega = A_1 \cup \dots \cup A_n$, a finite union of A_i 's, and hence in $\mathcal{D}$.

For $A \in \mathcal{D}$, since A is a finite union of A_i 's, $A = A_{i_1} \cup \dots \cup A_{i_k}$ for some $i_1, \dots, i_k$.

Now,

$$A^c = \bigcup_{i \in \{1, 2, \dots, n\} \setminus \{i_1, \dots, i_k\}} A_i \in \mathcal{D}.$$

Here keep in mind the following. When $A = \Omega$, then $\{i_1, i_2, \dots, i_k\} = \{1, 2, \dots, n\}$. Then

$$A^c = \bigcup_{i \in \{1, 2, \dots, n\} \setminus \{i_1, \dots, i_k\}} A_i$$

is the empty union of sets which is $\emptyset$.

Now for $A = A_{i_1} \cup \dots \cup A_{i_k}, B = A_{j_1} \cup \dots \cup A_{j_l} \in \mathcal{D}$,

$$A \cup B = A_{i_1} \cup \dots \cup A_{i_k} \cup A_{j_1} \cup \dots \cup A_{j_l} \in \mathcal{D}.$$

Hence $\mathcal{D}$ is a field and hence a σ -field by using Problem 1.

For checking 2., note that each A_i is a 'one' union of A_i and hence are in $\mathcal{D}$. This prove 2.

Let $\mathcal{G}$ be a σ -field containing $\{A_1, \dots, A_n\}$, then from the definition of σ -field, it contains all the finite union of A_i 's. Hence $\mathcal{D} \subseteq \mathcal{G}$. That's all.

Chapter 3 : Random Variables-General Facts

Key words: Random variable, Borel σ -field of subsets of $\mathbb{R}$, σ -field generated by random variable.

This chapter explains some general ideas related to random variables. In many situations, one is interested only in some aspects of a random experiment/phenomenon. For example, consider the experiment of tossing 3 unbiased coins and we are only interested in number of 'Heads' turned up. The probability space corresponding to the experiment of tossing 3 coins is given by

$$\Omega = \{(\omega_1, \omega_2, \omega_3) \mid \omega_i \in \{H, T\}\}, \mathcal{F} = \mathcal{P}(\Omega)$$

and P is described by

$$P(\{(\omega_1, \omega_2, \omega_3)\}) = \frac{1}{8}.$$

Our interest is in knowing no. of 'Heads', i.e., in the map $(\omega_1, \omega_2, \omega_3) \rightarrow I_{\{\omega_1=H\}} + I_{\{\omega_2=H\}} + I_{\{\omega_3=H\}}$. Here I_A denotes the 'indicator' function of A defined by

$$I_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A \\ 0 & \text{if } \omega \in A^c. \end{cases}$$

In general we are interested in functions of sample space. Also in many cases, one never observe directly the underlying random phenomenon but observe 'measurements' coming from it and these measurements can be modeled as functions of the sample space. So in short, mainly our interest lies in functions of sample space and its analysis in order to arrive at 'conclusions' about the random phenomenon. But to study a function associated with a random phenomenon, one should be able to assign probabilities to 'reasonably large class of events associated with the function. In general we can't do this for all functions defined on the sample space. So one need to restrict ourself to certain class of functions of the sample space for which we can do this and we call them random variables. In short, random variables are nothing but 'measurable observations' from a random phenomenon.

Definition 2.1 Let Ω be a sample space and $\mathcal{F}$ a σ -field of subsets of Ω . A function $X : \Omega \rightarrow \mathbb{R}$ is said to be a random variable (with respect to $\mathcal{F}$) if

$$\{\omega \in \Omega \mid X(\omega) \leq x\} \in \mathcal{F} \text{ for all } x \in \mathbb{R}.$$

Now on, we denote $\{\omega \in \Omega \mid X(\omega) \leq x\}$ by $\{X \leq x\}$. Also we suppress Ω and $\mathcal{F}$ from the description of X when Ω and $\mathcal{F}$ are understood from the

context, i.e. we just say that ' X is a random variable'

Remark 0.1 (1) In the definition of random variable, the probability measure P is not in the picture.

(2) Also it is possible that a function which is not be a random variable with respect to a σ -field becomes a random variable with respect to another σ -field.

Example 0.1 Let $\Omega = \{H, T\}$, $\mathcal{F} = \mathcal{P}(\Omega)$. Define $X : \Omega \rightarrow \mathbb{R}$ as follows:

$$X(H) = 1, X(T) = 0.$$

Then X is a random variable.

Note that any map $Z : \Omega \rightarrow \mathbb{R}$ is a random variable for the probability space given in Example 0.1.

Now, we give an example, where this is not the case.

Example 0.2 Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\mathcal{F} = \sigma(\{\{2\}, \{4\}, \{6\}, \{1, 3, 5\}\})$. Then $X(1) = 1, X(\omega) = 0$ otherwise is not a random variable with respect to $\mathcal{F}$. Note that X is a random variable with respect to the σ -field $\mathcal{P}(\Omega)$.

X is a random variable imply that the 'basic events' $X^{-1}((-\infty, x])$ associated with X are in $\mathcal{F}$. Does this imply a larger class of events associated with X are in $\mathcal{F}$ (i.e. can be assigned probabilities) ? To examine this, one need the following σ -field.

Definition 2.2 The σ -field generated by the collection of all open sets in $\mathbb{R}$ is called the **Borel σ -field** of subsets of $\mathbb{R}$ and is denoted by $\mathcal{B}_{\mathbb{R}}$. Also if $B \in \mathcal{B}_{\mathbb{R}}$, then B is said to be a **Borel set**.

Lemma 0.1 Let $\mathcal{I}_1 = \{(-\infty, x] \mid x \in \mathbb{R}\}$. Then $\sigma(\mathcal{I}_1) = \mathcal{B}_{\mathbb{R}}$

Proof. For $x \in \mathbb{R}$,

$$(-\infty, x] = (x, \infty)^c \in \mathcal{B}_{\mathbb{R}}.$$

Therefore

$$\mathcal{I}_1 \subseteq \mathcal{B}_{\mathbb{R}}.$$

Hence

$$\sigma(\mathcal{I}_1) \subseteq \mathcal{B}_{\mathbb{R}}. \tag{0.6}$$

Now we prove the reverse inclusion. For $b \in \mathbb{R}$,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n} \right] \in \sigma(\mathcal{I}_1).$$

Also, for $a, b \in \mathbb{R}$, $a < b$,

$$(a, b) = (-\infty, b) \setminus (-\infty, a] \in \sigma(\mathcal{I}_1).$$

Therefore, $\sigma(\mathcal{I}_1)$ contains all open intervals for the form (a, b) . Since any open set in $\mathbb{R}$ can be written as a (countable) union of open intervals with rational end points,¹ it follows that

$$\mathcal{O} \subseteq \sigma(\mathcal{I}_1),$$

where $\mathcal{O}$ denote the set of all open sets in $\mathbb{R}$. Thus,

$$\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{O}) \subseteq \sigma(\mathcal{I}_1). \quad (0.7)$$

Combining (0.6) and (0.7), we get

$$\sigma(\mathcal{I}_1) = \mathcal{B}_{\mathbb{R}}.$$

□

Theorem 0.2 *Let $X : \Omega \rightarrow \mathbb{R}$ be a function. Then X is a random variable defined on $(\Omega, \mathcal{F}, P)$ iff $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}_{\mathbb{R}}$.*

Proof. Suppose X is a random variable. i.e.,

$$\{X \leq x\} = X^{-1}\left((-\infty, x]\right) \in \mathcal{F} \text{ for all } x \in \mathbb{R}. \quad (0.8)$$

Set

$$\mathcal{D} = \{B \in \mathcal{B}_{\mathbb{R}} \mid X^{-1}(B) \in \mathcal{F}\}.$$

From (0.8), we have

$$\mathcal{I}_1 \subseteq \mathcal{D} \subseteq \mathcal{B}_{\mathbb{R}}. \quad (0.9)$$

¹Recall that from the definition of open set, given any element of the open set there is an open interval containing the element contained in the open set. Now using the property that 'between any two reals there is a rational' we can choose another open interval with rational end points containing the element and contained in the open set.

Now

$$\begin{aligned}
B \in \mathcal{D} &\Rightarrow X^{-1}(B) \in \mathcal{F} \\
&\Rightarrow [X^{-1}(B)]^c \in \mathcal{F} \\
&\Rightarrow X^{-1}(B^c) \in \mathcal{F} \text{ (since } [X^{-1}(B)]^c = X^{-1}(B^c)\text{)} \\
&\Rightarrow B^c \in \mathcal{D}.
\end{aligned}$$

Also

$$\begin{aligned}
B_1, B_2, \dots \in \mathcal{D} &\Rightarrow X^{-1}(B_n) \in \mathcal{F} \forall n = 1, 2, 3, \dots \\
&\Rightarrow \bigcup_{n=1}^{\infty} X^{-1}(B_n) \in \mathcal{F} \\
&\Rightarrow X^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right) \in \mathcal{F} \\
&\Rightarrow \bigcup_{n=1}^{\infty} B_n \in \mathcal{D}.
\end{aligned}$$

Hence $\mathcal{D}$ is a σ -field. Now from (0.9) and Lemma 0.1, it follows that $\mathcal{D} = \mathcal{B}_{\mathbb{R}}$. i.e, $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}_{\mathbb{R}}$.

Converse statement follows from the observation

$$\mathcal{I}_1 \subseteq \mathcal{B}_{\mathbb{R}}.$$

□

Remark 0.2 Theorem 0.2 tells that if $X^{-1}\left((-\infty, x]\right) \in \mathcal{F}$ for all $x \in \mathbb{R}$, then $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}_{\mathbb{R}}$.

Example 0.3 Let $\Omega = (0, 1]$, $\mathcal{F} = \mathcal{B}_{(0,1]}$. Define $X(\omega) = 3\omega + 1$. Here $\mathcal{B}_{(0,1]}$ is the σ -field generated by all open sets in $(0, 1]^2$
For $x \in \mathbb{R}$

$$\begin{aligned}
\{X \leq x\} &= \{\omega \in \Omega \mid 3\omega + 1 \leq x\}. \\
&= \begin{cases} \emptyset & \text{if } x < 1 \\ (0, \frac{x}{3} - \frac{1}{3}] & \text{if } 1 \leq x \leq 4 \\ (0, 1] & \text{if } x > 4. \end{cases}
\end{aligned}$$

Since

$$\emptyset, \left(0, \frac{x}{3} - \frac{1}{3}\right], (0, 1] \in \mathcal{B}_{(0,1]},$$

X is a random variable.

Lemma 0.2 For a non empty set Ω and a map $X : \Omega \rightarrow \mathbb{R}$, define

$$\sigma(X) = \{X^{-1}(B) \mid B \in \mathcal{B}_{\mathbb{R}}\}.$$

Then $\sigma(X)$ is a σ -field.

²A set O in $(0, 1]$ is open if $O = (0, 1] \cap G$, for some G which is an open set in $\mathbb{R}$.

Proof.

$$X^{-1}(\mathbb{R}) = \Omega.$$

Hence $\Omega \in \sigma(X)$.

For $A \in \sigma(X)$, there exists $B \in \mathcal{B}_{\mathbb{R}}$ such that

$$X^{-1}(B) = A \text{ and } A^c = X^{-1}(B^c).$$

From these, it follows that $A \in \sigma(X) \Rightarrow A^c \in \sigma(X)$. Similarly from

$$X^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right) = \bigcup_{n=1}^{\infty} X^{-1}(B_n)$$

it follows that

$$A_1, A_2, \dots \in \sigma(X) \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \sigma(X).$$

□

Definition 0.1 For a non empty set Ω and a map $X : \Omega \rightarrow \mathbb{R}$, $\sigma(X)$ is called the σ -field generated by X .

Remark 0.3 (1) The occurrence or non occurrence of the event $X^{-1}(B)$ tells respectively whether any realization $X(\omega)$ is in B or not. Thus $\sigma(X)$ collects all such information. Hence $\sigma(X)$ can be viewed as the information available with the random variable.

(2) For a function X defined on a sample space Ω , $\sigma(X)$ is a σ -field of subsets of Ω and is the smallest σ -field under which X is a random variable. In otherwords, a map $X : \Omega \rightarrow \mathbb{R}$ is a random variable with respect to a σ -field $\mathcal{F}$ iff $\sigma(X) \subseteq \mathcal{F}$.

Example 0.4 Let X be as in Example 0.2, i.e. $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $X = I_{\{1\}}$. Then for $B \in \mathcal{B}_{\mathbb{R}}$,

$$X^{-1}(B) = \begin{cases} \emptyset & \text{if } B \cap \{0, 1\} = \emptyset \\ \{2, 3, 4, 5, 6\} & \text{if } B \cap \{0, 1\} = \{0\} \\ \{1\} & \text{if } B \cap \{0, 1\} = \{1\} \\ \Omega & \text{if } B \cap \{0, 1\} = \{0, 1\} \end{cases}$$

Hence $\sigma(X) = \sigma(\{1\})$. It is easy to see that X is a random variable with respect to any σ -field containing $\sigma(\{1\})$ and $\sigma(\{\{2\}, \{4\}, \{6\}, \{1, 3, 5\}\})$, the σ -field in Example 0.2, doesn't contain $\sigma(\{1\})$.

Given two random variables X, Y on $(\Omega, \mathcal{F})$,³ and $c \in \mathbb{R}$, we define sum, scalar product and product of X, Y , i.e. $X + Y : \Omega \rightarrow \mathbb{R}, cX : \Omega \rightarrow \mathbb{R}, XY : \Omega \rightarrow \mathbb{R}$ as follows.

$$(X+Y)(\omega) = X(\omega)+Y(\omega), (cX)(\omega) = cX(\omega), (XY)(\omega) = X(\omega)Y(\omega), \omega \in \Omega.$$

Theorem 0.3 *If X, Y are random variables on $(\Omega, \mathcal{F})$ and $c \in \mathbb{R}$, then $X + c, cX, X + Y, X^2$ and XY are random variables with respect to $\mathcal{F}$.*

Proof: The proof of first two are simple exercises.

For $a \in \mathbb{R}$,

$$\{X + Y < a\} = \bigcup_{r \in \mathbb{Q}} (\{X < r\} \cap \{Y < a - r\}) \in \mathcal{F}.$$

The proof of the above identity, only uses "between any two real numbers, there is always a rational"⁴ Now

$$\{X + Y \leq a\} = \bigcap_{n=1}^{\infty} \left\{ X + Y < a + \frac{1}{n} \right\}.$$

Hence $\{X + Y \leq a\} \in \mathcal{F}$ for all $a \in \mathbb{R}$. Therefore, $X + Y$ is a random variable.

For $a \in \mathbb{R}$,

$$\{X^2 \leq a\} = \begin{cases} \emptyset \in \mathcal{F} & \text{if } a < 0 \\ \{X = 0\} = X^{-1}(\{0\}) \in \mathcal{F} & \text{if } a = 0 \\ \{-\sqrt{a} \leq X \leq \sqrt{a}\} = X^{-1}([-\sqrt{a}, \sqrt{a}]) \in \mathcal{F} & \text{if } a > 0. \end{cases}$$

(In the above $\{0\}$ and $[-\sqrt{a}, \sqrt{a}]$ are closed sets and hence Borel sets.) Hence, X^2 is a random variable.

Note that

$$XY = \frac{1}{2}[(X + Y)^2 - X^2 - Y^2].$$

Since $\frac{1}{2}(X + Y)^2, -\frac{1}{2}X^2, -\frac{1}{2}Y^2$ are random variables and XY is their sum, XY is a random variable.

³we say that X is a random variable on $(\Omega, \mathcal{F})$ if $X : \Omega \rightarrow \mathbb{R}$ and is a random variable with respect to the σ -field $\mathcal{F}$

⁴Hint: For $X(\omega) + Y(\omega) < a$, choose r , a rational number such that $X(\omega) < r < a - Y(\omega)$. Then $\omega \in \{X < r\} \cap \{Y < a - r\}$.

Theorem 0.4 (i) If X, Y are random variables with respect to $\mathcal{F}$, so are $\min\{X, Y\}, \max\{X, Y\}$.

(ii) If $\{X_n\}$ is a sequence of random variables on $(\Omega, \mathcal{F})$ which is bounded from above and

$$X(\omega) = \sup_n X_n(\omega), \text{ for all } \omega \in \Omega.$$

Then X is a random variable with respect to $\mathcal{F}$. Analogous statement is true for infimum.

Proof:

(i) Set $Z = \min\{X, Y\}$. For $a \in \mathbb{R}$,

$$\{Z \leq a\} = \{X \leq a\} \cup \{Y \leq a\} \in \mathcal{F}.$$

Therefore Z is a random variable.

The proof of $\max\{X, Y\}$ is a random variable is similar.

Proof of (ii) follows from

$$\{X \leq a\} = \cap_{n=1}^{\infty} \{X_n \leq a\}.$$

□